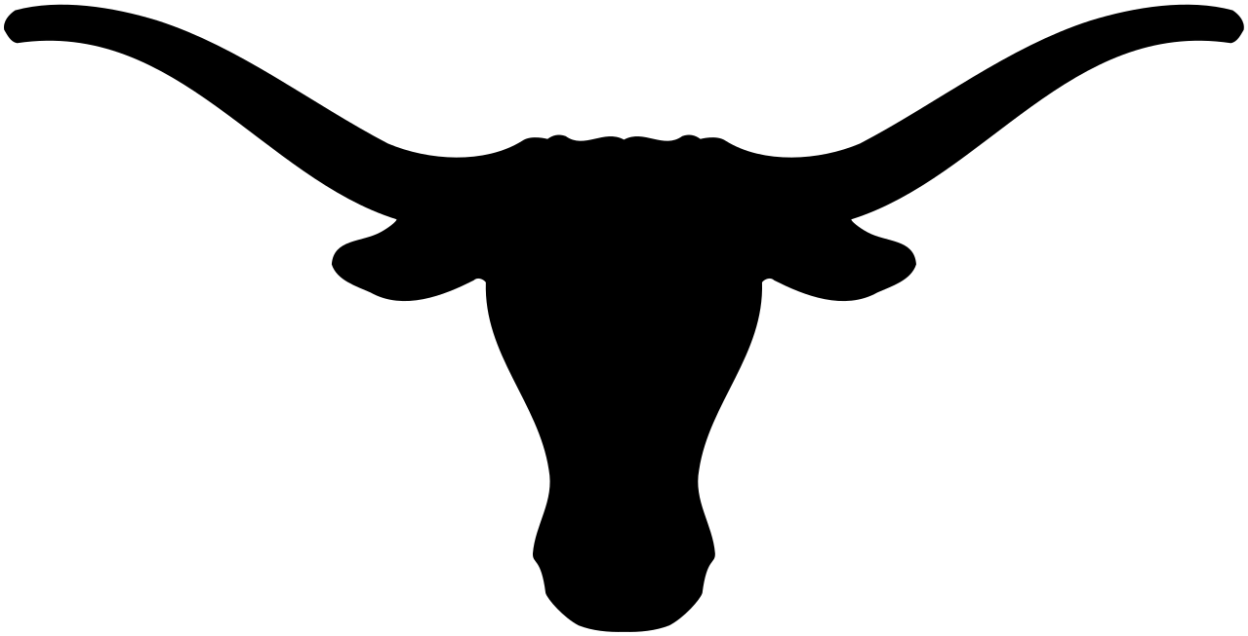
UT Regional 2022



Dynamic Planet

Division B/C

Test Booklet

CLASS SET : DO NOT WRITE ON THIS

1. What percentage of Earth's water is freshwater? (1 pt)
    a. 2.5%
    b. 5%
    c. 1%
    d. 10%
2. How much would sea level rise if all glaciers melted? (1 pt)
    a. 2 meters
    b. 70 meters
    c. 500 meters
    d. 25 meters
3. Which of the following has the longest residence time? (1 pt)
    a. Deep Groundwater
    b. Valley Glaciers
    c. Ice Caps
    d. Atmosphere
4. Assume the famed Zilker Park (a relatively flat landscape) has altering patterns of weak and resistant bedrock. If a system of streams were to form on this landscape, what sort of drainage pattern would it be? (1 pt)
    a. Dendritic
    b. Rectangular
    c. Radial
    d. Trellis
5. Put the following three in order of occurrence in the formation of a river. (2 pt)
    a. Main Channel
    b. Sheetwash
    c. Tributaries

6. What is the proper term for the lengthening of a stream channel by erosion at the origin of the stream? (1 pt)
    a. Mouth Widening
    b. Headward Erosion
    c. Downcutting
    d. Head Expansion

7. How can groundwater sapping increase the size of a stream's origin? (2 pt)

8. Which of the following locations would most likely have a rectangular drainage pattern? (1 pt)

a. Badlands
b. Highly-jointed plain
c. A gentle hill
d. Flat parking-lot

9. What separates watersheds from each other? (1 pt)

a. Natural levees
b. Drainage divide
c. Drumlins
d. Rills

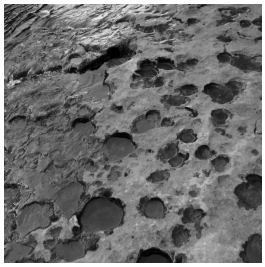
10. Are all losing streams also ephemeral streams? Explain your answer. (2 pt)

11. If one were to measure the residence time of the main channel, they would notice that it is actually shorter than the residence time of the entire river. One reason for this is the presence of the hyporheic zone. Explain how the hyporheic zone increases the residence time of the entire stream. (3 pt)

12. There is a river that is the shape of a half-circle. If 500 cubic meters of water passes through the river every 5 seconds, and the rate of water flow is 25 m/s, what's the radius of the river? (3 pt)

13. Assume you have 2 identical rivers that have the same cross sectional area. However, river A is a straight, semicircular channel, and river B has a wide, shallow channel. Which of the two rivers would be the safest to cross, and explain your answer. Make sure to mention velocity variations in your answer. (3 pt)

14. Shown below is an image of a very common type of river formation. Explain how it formed. (2 pt)



15. What percentage of the gravitational potential energy stored in a river transforms into kinetic energy used to erode the walls and beds of a channel? (1 pt)

a. 15%
b. 1%
c. 3%
d. 25%

Complete the following sentences:

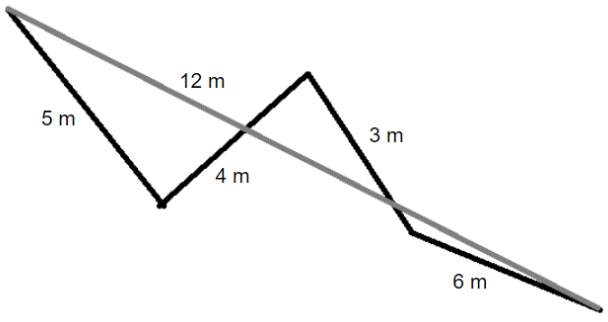
16. Bed load movement commonly occurs through _____. (1 pt)
17. As competence increases, _____ particles can be carried. (1 pt)
18. A stream's capacity depends on _____ and _____. (1 pt)

19. The Mississippi River has the nickname "Big Muddy" for the large amount of clay and silt that it carries. What type of sediment load is this? (1 pt)

a. Bed load
b. Dissolved load
c. Suspended load
d. Carried load

20. For the following scenarios, state if the competence of the river would increase or decrease.

a. Stream becomes wider. _____ (0.5 pt)
b. Stream decreases in speed._____ (0.5 pt)
c. Sudden flood greatly increases discharge of the river. _____ (0.5 pt)
d. Uplift occurs at the base of the stream. _____ (0.5 pt)



21. Find the sinuosity of the above river. (2 pt)

22. What class of sinuosity is this? (1 pt)



23. What is the name of the type of topography to the left? (1 pt)

24. Looking off into the landscape, you notice that it is riddled with holes, caves, and underground rivers. Explain how the landscape to the left formed. (3 pt)

25. What is the material that makes up a majority component of this type of landscape? (1 pt)

26. Write out the chemical equation that leads to the creation of the weak base that dissolves the limestone. (2 pt)



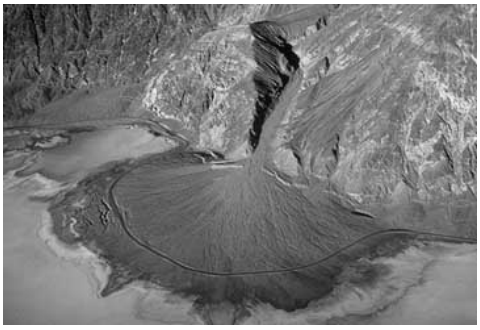
Shown to the left is a meandering stream.

27. At which location (A, B, or C), is the river bed the deepest, and why? (2 pt)



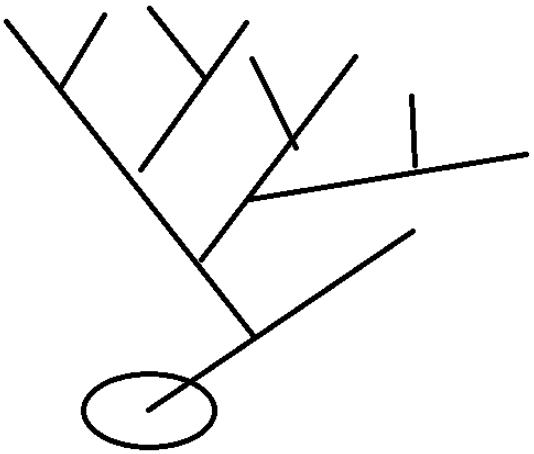
28. Sometimes, an oxbow lake like the one shown to the right forms from a meandering river. Explain the formation of an oxbow lake. (3 pt)

29. Throughout the life of a meandering stream, its meander becomes more and more intense through a positive feedback loop. Explain the process of this feedback loop. (2 pt)



30. What is the minimum sinuosity for a river to be classified as a meandering stream?(1 pt)

31. What is the name of the landform to the left, and how did it form? (2 pt)



Take the following stream to the left:

32. Find the maximum classic, strahler, and shreve stream orders of the river. (3 pt)

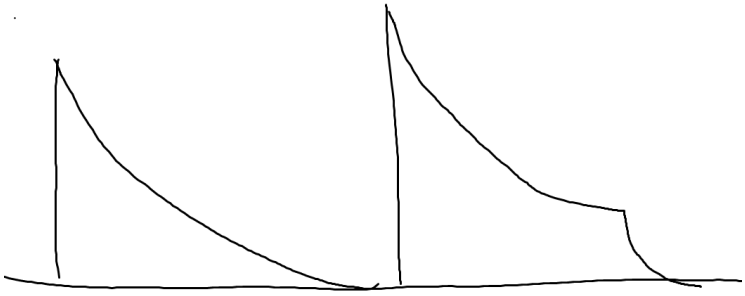
33. What is the maximum stream order of the Amazon river? (0.5 pt)

34. Many times, along the edge of a meandering stream, there is a little lip of sediment known as a natural levee. Explain how a natural levee forms. (2 pt)

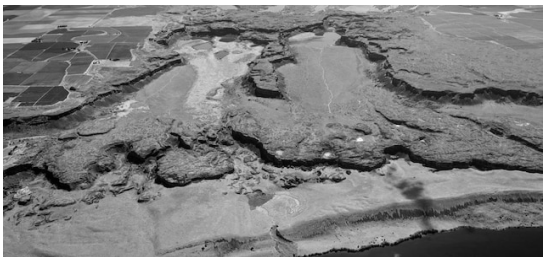
35. What occurs to tributaries when natural levees become too large? What are these tributaries called?(2 pt)

36. What shape would a delta formed from a river whose current exceeds that of the ocean take?(2 pt)

37. Shown below is a diagram of two river profiles. The profile on the left is an ideal river profile; however, the one on the right is representative of a more realistic profile. What might cause the little lip on the right profile, and what is the name of that location? (1 pt)



38. In some localities, a river switches between normal and flood flow on a regular schedule. However, the two different flows have different competences, so when they reach a flat plain, they end up creating what type of river? (1 pt)



Shown to the left is a famous landscape in eastern Washington called the channeled scablands.

39. Explain how glacial processes lead to the creation of this landscape. (2 pt)

40. State whether the following river characteristics increase or decrease as the water approaches the mouth of the river.

a. Channel roughness _____(0.5 pt)
b. Discharge _____(0.5 pt)
c. Channel size _____(0.5 pt)
d. Channel slope _____(0.5 pt)
e. Flow Velocity _____(0.5 pt)

41. What percentage of the earth's liquid freshwater is groundwater? (1 pt)

A. 54%
B. 32%
C. 5%
D. 96%

42. When water infiltrates into the ground, a fair amount gets stuck at the surface creating a belt of soil moisture. What is the reason for this? (1 pt)

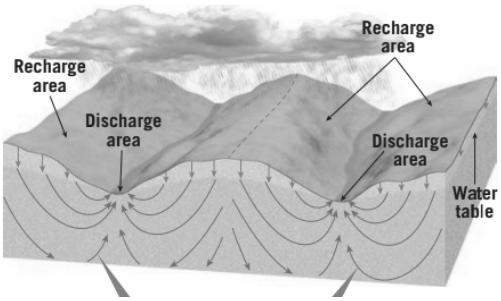
A. Small pore size traps the water
B. Molecular attraction holds it in place
C. Roots attract the water and hold it
D. Large pore sizes prevent water movement

43. Why is it possible to pump water from the zone of saturation, but not from an unsaturated zone even though both may have large amounts of water? (2 pt)

44. How does the fact that groundwater moves very slowly affect the shape of the water table? (2 pt)

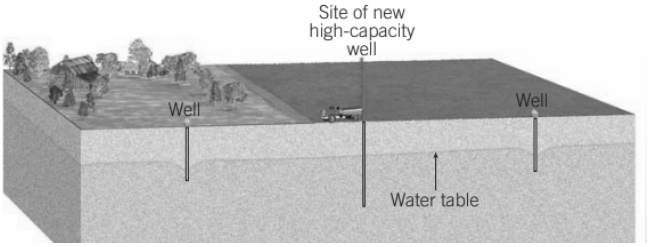
45. The two factors in subsurface materials that influence the movement of groundwater are ___???____ and _____???_____. (2 pt)

46. In sedimentary rocks, shape/size of the grains, degree of sorting, packing factor, and many other factors affect pore size of the rock. However, metamorphic and igneous rocks lack these structures. Instead, what must provide the voids for water to flow through? (2 pt)



47. Shown to the left is an image of infiltration and the water table. Notice how the water flows downwards before turning and flowing upwards. Explain how this happens. (3 pt)

48. Shown to the right is an image of a farm with 3 wells. What would happen to all the wells if the middle well were to excessively draw water? (3 pt)



49. If sewage enters groundwater, it may be purified by natural processes. However, this depends on having the right type of sediment. Shown below are three different types of aquifers to choose from. Choose the one that is best for filtering out sewage, and then explain why it is correct. Also explain why the other two are incorrect. (4 pt)

A. Coarse gravel aquifers
B. Sand based aquifers
C. Clay based aquifers

50. Why are most caverns created at or below the water table? (2 pt)

Shown below is an image of a very well-known karst landscape in China.



51. Explain the formation of this unique landscape, and why they generally occur more rapidly in tropical climates. (2 pt)